

Leveraging AlphaEarth Foundations Embeddings for High-Accuracy County-Scale Corn and Soybean Yield Estimation

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Accurate, timely crop yield estimates are critical for food security and agricultural planning, yet most operational pipelines still rely on handcrafted indices or weather-driven regressors. We assess whether Google DeepMind's AlphaEarth Foundations (AEF), a multimodal Geo-Foundation Model that provides annual 64-D embeddings at 10 m resolution, are sufficient for county-level U.S. corn yield estimation. Using one embedding per county-year and a light RBF-SVR learner, we evaluate leave-one-year-out generalization. Despite the absence of time-resolved inputs, the embeddings yield strong predictive accuracy across years. For corn, mean test performance is $R^2 = 0.825$ with RMSE 14.8 bu/ac, reaching $R^2 = 0.861$ in the best year. The same workflow transfers to soybean with $R^2 = 0.814$. Spatial diagnostics show strong positive autocorrelation of residuals every year, indicating missing slow-varying drivers rather than stochastic noise. Test-fold permutation importance shows feature importance, with several directions shared across crops, while a few flip sign, suggesting crop-specific semantics. These results demonstrate that annual AEF embeddings serve as analysis-ready features for county-scale yield estimation, enabling a lightweight, reproducible pipeline (mask, mean, regress) that is competitive with more complex alternatives.

Keywords

alphaearth, corn, county-level, crop yield estimation, geo-foundation models, geoai, geoscience, remote sensing

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Index Terms—Crop yield estimation, GeoAI, Geo-Foundation Models, AlphaEarth, Corn, Remote sensing, County-level.

I. INTRODUCTION

Earth observation (EO) has entered the big-data era, with petabytes of multisensor imagery becoming freely available each year. Translating such amount of heterogeneous data streams into actionable knowledge remains challenging, as most downstream applications still rely on task-specific feature engineering and bespoke models. Inspired by the success of large “foundation models” in various fields across the natural and social sciences, the EO community is now developing Geo-Foundation Models (GFMs), which are large, self-supervised networks trained on globally distributed satellite data to learn reusable spatio-temporal representations of Earth’s surface [1]–[4]. These models promise to reduce the cost of labeled data, accelerate model development, and standardize EO analytics across geographies and tasks.

Early efforts to build GFMs such as Seasonal Contrast (SeCo) [5] and Geography-Aware Self-Supervised Learning [6] focused on learning universal visual features from satellite imagery while incorporating spatial information, laying the groundwork for large-scale, task-agnostic representation learning in EO [3], [7]. Building on this vision-only foundation, researchers quickly expanded to richer spectral information, such as SatMAE [8], SpectralEarth [9], SpectralGPT [10], which show state-of-the-art performance on EO specific classification and retrieval tasks. These advances signal a decisive move away from handcrafted features toward compact, high-capacity embeddings that readily integrate multispectral context. The next trend is both multisensor and multi-temporal. Prithvi-EO 1.0 [11] incorporates multi-temporal and multi-spectral data while its 2.0 version demonstrates an adaption with multi-modal data for downstream tasks [12].

Google DeepMind’s recent AlphaEarth Foundations (AEF) pushes the paradigm further, fusing multi-year, multi-modal satellite data into a 64-dimensional unit-sphere embedding for every 10 m land pixel worldwide from 2017 onward [13]. Early results show that these temporally resolved, multi-modal representations enable state-of-the-art performance in unsupervised land-cover mapping, change detection, and similarity search—often with little or no additional network training [13]. Collectively, these developments mark a paradigm shift toward scalable, sensor-agnostic Geo-Foundation Models whose unified embeddings can be analysis-ready for various EO applications.

A long-standing testbed of EO analytics is crop-yield prediction, particularly for the U.S. Corn Belt where county-level statistics for corn and soybean have been recorded since the early 1900s [14]. In the early stages of research, process-based models (PBMs) were widely employed in this field. These models utilize real-world environmental parameters to simulate the physical processes underlying crop growth, thereby providing strong interpretability. For example, [15] applied Decision Support System for Agrotechnology Transfer (DSSAT)—one of the most widely used crop models—to estimate corn yield fluctuations by running simulations across 51 weather stations. Although PBMs offer promising performance and interpretability, they often require specific types of input data and expert calibration to achieve reliable results [16]. As technology has advanced, machine learning (ML)

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methods have become increasingly prevalent in this field. Unlike PBMs, ML models establish empirical relationships between input parameters and target variables, offering greater flexibility in application and reducing the need for extensive expert knowledge and calibration. For instance, [17] employed a Random Forest algorithm with MODIS satellite imagery to forecast corn and soybean yields, achieving a mean absolute error (MAE) of 0.89 Mg/ha. Similarly, [18] evaluated multiple ML models using MODIS-derived spectral indices to predict county-level corn yields in the U.S. Corn Belt, with the best-performing model achieving an R^2 of 0.67.

In recent years, advanced deep learning (DL) models have attracted increasing attention in crop yield prediction. Ranging from convolutional neural networks (CNNs) to Transformer-based architectures, many SOTA models have been applied in this field, achieving not only higher prediction accuracy but also enhanced functionality. For example, [19] employed the Bayesian Neural Network (BNN) to incorporate uncertainty estimation, thereby improving model interpretability. [20] proposed DeepCropNet, a hierarchical architecture for time-series corn yield estimation, achieving a RMSE of 0.86 Mg/ha. [21] adopted the multiple instance learning (MIL) framework, treating each county as a “bag” of pixels and assigning attention weights to individual pixels to improve spatial feature utilization. Despite these advancements, DL models in crop yield prediction still face challenges such as data scarcity, domain shift across regions or years, and the need for better integration of domain knowledge.

To our knowledge, this is one of the earliest evaluation of AlphaEarth Foundations for a real-world downstream task—county-scale yield estimation. We evaluate whether a Geo-Foundation model’s embeddings, namely, AlphaEarth Foundations’ 64-band annual vectors, can serve as off-the-shelf features for county-scale yield estimation in the U.S. Corn Belt. We hypothesize that the embeddings implicitly encode phenological dynamics, soil moisture and climatic variability over a year’s time window, thereby providing a richer, more transferrable signal than many handcrafted indices. By coupling these representations with contemporary regression ensembles, we aim to answer on following key questions that arise when applying AlphaEarth Foundations’ embedding to real-world Earth Observation tasks:

- 1) **Predictive power.** How well do embedding-only models generalize across years for corn (and soybean)?
- 2) **Dimensional parsimony.** How many embedding dimensions are needed to retain most out-of-sample skill?
- 3) **Robustness.** How stable are errors across space and time, and do residuals exhibit interpretable spatial structure?

II. DATA AND METHODS

A. AlphaEarth Foundations Embedding Dataset

Google’s AlphaEarth Foundations model generates a 64-dimensional embedding vector for every 10 m land pixel by jointly encoding annual time-series from sources including optical images, radar and auxiliary climatic inputs [13]. Since each component is a coordinate on a 64-D unit-sphere, the vector can be treated like a 64-band “spectral signature” that is sensor-agnostic, spatially explicit and temporally regular over a year. Annual composites are provided through Google Earth Engine (GEE, [22]) as the image collection covering calendar years 2017–2024. We use data for all eight available years.

Our region of interest is the U.S. Corn Belt states, namely, Illinois, Indiana, Iowa, Kansas, Kentucky, Michigan, Minnesota, Missouri, Nebraska, North Dakota, Ohio, South Dakota and Wisconsin, which account most of the corn production in the United States [18]. County-level data were clipped with county boundaries in GEE platform. Within each county we generated crop-specific masks using the USDA Cropland Data Layer (CDL) for the corresponding year. We mainly utilized CDL class 1 (corn) data [23] while we also generated class 5 (soybean) data to test our pipeline’s performance and transferability. The crop-masks were then applied to mask the embedding layer for aggregated (county-level mean) statistics. All preprocessing was executed in GEE to avoid local storage of terabytes rasters. The exported feature matrix therefore contains 64 predictors per county-year-crop record and is fully reproducible with the public GEE script provided in supplementary materials.

B. County-level Yield Labels

County-level reported yield data were retrieved from the USDA National Agricultural Statistics Service (NASS) Quick Stats database through its REST API (<https://quickstats.nass.usda.gov/api>). Our queries specified county-level, corn/soybean yield data. Only survey-based, end-of-season estimates were used to ensure methodological consistency.

After filtering, the analysis set contained 6,350 county-year records from 992 counties covering the U.S. Corn Belt states. The county coverage with the years of available data are shown in Fig 1. The 64 embedding dimensions (A00-A63) constituted the sole predictor set.

C. Predictive Modeling Framework

For the model sweep (Table I) we compared Linear Regression, Support Vector Regression (SVR), Linear SVR, Lasso, Ridge, Elastic Net, Random Forest, Gradient Boosting, MLP, and k -NN. Based on benchmarking and parameter sweeping results, we selected an SVR with a radial basis kernel (RBF) as our primary estimator. All subsequent metrics and statistics are based on the final models results.

SVR is a regularized, data-efficient method that works well in high-dimensional settings [24], [25]. The RBF

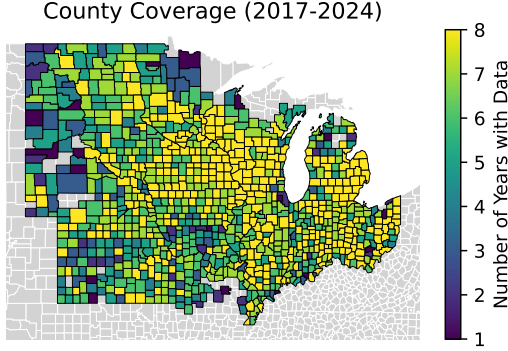


Fig. 1. Counties (Corn Belt) and years with available labels and AEF coverage (2017-2024).

kernel is flexible and broadly competitive [26]–[28], with mature implementations available [29]. Together, these make RBF-SVR a simple, strong choice here without further ablations.

We evaluate the models using leave-one-year-out (LOYO) cross-validation, which is commonly applied in crop yield estimation, particularly when data availability is limited [30]. In this approach, one calendar year is withheld as the test set while the remaining years are used for training, differing from the chronological train-test split that trains on earlier years and tests on later ones [31]. Although LOYO does not fully replicate real-world forecasting—since later years may be used to predict earlier years—it maximizes the use of available observations [32] and provides a more balanced evaluation of model performance across years [18]. This “group-by-year” blocking prevents within-year information leakage and yields an unbiased estimate of performance under out-of-year covariate shift (e.g., weather anomalies, management changes). As a result, the term “prediction” used in this paper, is different from “forecasting” [33]. The selected model is fitted on the full training-years data before scoring the target year.

Hyperparameters were tuned once via grid search on training years and then fixed for all LOYO evaluations. The selected configuration is $C \approx 52$, $\gamma \approx 0.0070$ and $\epsilon \approx 0.99$. Generalization across time is evaluated with LOYO over 2017-2024. We report per-year R^2 , Root Mean Square Error (RMSE), and averaged metrics over all year leave-out predictions as overall indicator of model performance for comparison with literature reports.

As a compact phenology comparator, we implemented a classical baseline, including: MODIS NDVI seasonal summaries over DOY 100–300 (mean, max, integral, and DOY of max) from MOD13Q1 (250 m, 16-day), PRISM monthly precipitation (12 bands, mm), and growing/cooling degree-day totals (GDD base 10 °C capped at 30 °C; CDD base 18 °C) using PRISM daily aggregates. All layers are crop-masked and averaged to county following the same LOYO protocol as AEF, we train Linear, Ridge, Random Forest, Gradient Boosting, and RBF-SVR; per-year performance is reported for the best baseline model.

In addition, we include county-mean baseline: predict each test county’s yield by the training-years mean yield of that county. Counties lacking a training mean in a given fold are omitted from that fold’s test set.

We also report roll-forward forecasting (train on $\leq t-1$, test on t) for 2019-2024 to assess operational realism.

D. Spatial analysis of error and model bias

To test whether errors are spatially structured, we compute global Moran’s I on the county-level relative errors in each test year. County centroids define a k -nearest neighbor graph with $k=8$. We report the normal deviate (z) and a permutation p -value (999 permutations) for significance. We compute Local Indicators of Spatial Association (LISA, [34]) on the same W to identify hot/cold spots. Counties are labeled HH (high error surrounded by high), LL (low surrounded by low), HL, or LH using $\alpha=0.05$. The per-year LISA maps visualize where over- or under-prediction clusters.

E. Ablation and Model Explainability

To relate the geometry of the AEF representation to model behavior, we analyze principal components (PCs) of the embeddings. For each held-out year, we fit PCA on standardized embeddings from the training years, project the test-year counties into that space without refitting, and (i) report the training-fold scree (variance explained by each PC in the embeddings), (ii) compute univariate test R^2 between each PC score and the model’s error (Absolute Error) within the test year, and (iii) fit a linear model of error on the first k PCs to obtain cumulative test R^2 as k increases. This procedure isolates whether residual errors align with the dominant unsupervised axes of variation.

we compute test-fold permutation importance to quantify how much each AEF dimension contributes to out-of-sample performance of the final model [35]. For a given held-out year, we measure the drop in R^2 when shuffling one feature column of the test set while leaving the trained model unchanged; we repeat the shuffling 20 times and average to reduce Monte-Carlo variance. This yields a per-year importance $\Delta R_{y,j}^2$ for feature j ; we summarize across years by reporting the mean and standard deviation of $\Delta R_{y,j}^2$. As a light-weight indicator of directionality, we also compute the Pearson correlation between each feature and the model’s *out-of-sample* predictions pooled over all folds; positive (negative) values indicate that larger values of that embedding dimension tend to increase (decrease) predicted yield on average. We emphasize that permutation importance is computed on held-out data only and is therefore reflective of generalization rather than in-sample fit.

III. RESULTS

A. County-Level Corn Yield Estimation

We first benchmarked a set of out-of-the-box regression models from `scikit-learn` package; the average leave-one-year-out test performance is summarized in Table I.

Model	Mean Test R^2
LinearRegression	0.6968
Ridge	0.6980
Lasso	0.6974
ElasticNet	0.7156
SVR_rbf	0.8077
LinearSVR	0.7223
RandomForest	0.7695
GradientBoosting	0.7546
MLPRegressor	0.6761
KNeighborsRegressor	0.7140

TABLE I

LEAVE-ONE-YEAR-OUT TEST (CORN): MEAN TEST R^2 BY MODEL, THE BENCHMARK USE UNTUNED MODEL SO THE BEST RESULT IS DIFFERENT THAN THE FINAL ONE WE USED.

Nearly all models achieve moderate to strong accuracy; even linear baselines reach a satisfactory coefficient of determination at $R^2 \approx 0.70$. Tree-based ensembles are consistently strong, e.g., RandomForest (0.770). The best overall is SVR with an RBF kernel (0.808), indicating that smooth nonlinear effects are important [36]. Overall, models that capture modest *nonlinearity* with effective regularization generalize best across years.

We then ran extensive parameter sweeping on SVR-RBF model, which gives us best-performing hyperparameters of as described before. The predictive skill, summarized in Table II, consistently demonstrates high accuracy. The R^2 across the eight independent test years averaged 0.825, with individual year values ranging from 0.744 (2017) to 0.861 (2021). The scatter plot (Fig. 2) shows individual Prediction vs. Actual yield relationship. The average RMSE of 14.78 bushels per acre (bu/ac) reflects a prediction error under 9% relative to mean county yields, representing an agronomically acceptable level of uncertainty. The result is close or even above many recent works on county-level corn yield prediction with temporal vegetation indices [19], [21], [36], [37]

Comparing to baselines, although training year-mean prediction is strong at the county scale (mean $R^2 \approx 0.69$ for both crops), AEF+SVR improves consistently. On corn, AEF reduces RMSE by 23.4% on average and by 21.8% on soybean. The best per-year NDVI+PRISM+GDD/CDD model averages $R^2=0.687$ on corn and 0.732 on soybean, while AEF reaches 0.825 and 0.814, respectively. Per-year margins for corn are large and uniform (e.g., +0.197 in 2018, +0.173 in 2019, +0.216 in 2020). The results indicate that AEF captures inter-annual anomaly structure beyond fixed county effects and seasonal NDVI/precipitation summaries.

Under the forecasting-style roll-forward protocol (train on past years only), AEF+SVR retains high accuracy across 2019–2024 with mean test $R^2=0.824 \pm 0.027$ and mean RMSE 14.88 bu/ac. Compared to LOYO, the year-by-year differences are small (mean $\Delta R^2=-0.011$), indicating that performance *does not* depend on access to future-year data. The mean signed relative error over 2019–2024 is +0.26% (bias near zero) and the error standard deviation is 14.77 bu/ac, consistent with the observed RMSE.

TABLE II

COUNTY-LEVEL YIELD ESTIMATION UNDER LOYO (2017–2024). EACH SUBTABLE REPORTS PER-YEAR TEST R^2 AND RMSE (BU/AC) FOR (I) AEF+SVR (THIS STUDY), (II) COUNTY-MEAN, AND (III) NDVI+PRISM+GDD/CDD BASELINE (BEST PER-YEAR MODEL).

Year	AEF+SVR		County mean		NDVI+W X (best)	
	R^2	RMSE	R^2	RMSE	R^2	RMSE
2017	0.744	17.12	0.766	16.29	0.735	15.70
2018	0.844	14.35	0.726	19.01	0.647	19.36
2019	0.810	12.59	0.513	20.13	0.637	15.83
2020	0.855	11.98	0.682	17.68	0.639	17.70
2021	0.861	15.20	0.709	21.74	0.750	20.05
2022	0.851	16.09	0.760	20.44	0.745	20.11
2023	0.838	14.96	0.724	19.50	0.714	18.75
2024	0.799	15.94	0.655	20.88	0.627	19.57
Mean	0.825	14.78	0.692	19.46	0.687	18.76

(a) Corn

Year	AEF+SVR		County mean		NDVI+W X (best)	
	R^2	RMSE	R^2	RMSE	R^2	RMSE
2017	0.789	4.33	0.707	5.10	0.765	4.48
2018	0.772	4.83	0.716	5.40	0.642	5.84
2019	0.717	4.57	0.387	6.73	0.701	4.55
2020	0.841	3.96	0.711	5.20	0.728	4.87
2021	0.863	4.88	0.737	6.78	0.801	5.86
2022	0.868	4.65	0.740	6.54	0.790	5.82
2023	0.832	5.10	0.731	6.45	0.673	7.02
2024	0.828	4.76	0.764	5.59	0.759	5.37
Mean	0.814	4.64	0.687	5.97	0.732	5.49

(b) Soybean

B. Spatial Analysis of Estimation Error

Fig. 3a annotates the cross-sectional relative error in each leave-one-year-out test. Visual inspection shows coherent swaths of under-prediction (blue) and over-prediction (red), rather than random speckle, suggesting structured residuals that are consistent with agro-ecological patterns. Prediction error also shows small but non-zero inter-annual bias that changes sign across years. In all cases the magnitude is within $\pm 5\%$, indicating a modest year-level intercept shift rather than a large-scale miscalibration. Moran’s I indicates moderate to strong positive spatial autocorrelation of the relative errors (Table IV). Values range from $I=0.3171$ (2024) to $I=0.4908$ (2021). All years are highly significant with large normal deviates ($z=16.20$ – 28.77) and permutation $p=0.001$, meaning neighboring counties tend to share the same sign and magnitude of error.

Local clustering (Fig. 3b) reveals contiguous HH (over-prediction) and LL (under-prediction) zones that recur across multiple years. For example, we observe repeated HH clusters along portions of the northern tier in several years and LL clusters across parts of the central Corn Belt in others. Mixed HL/LH edge patterns appear along transitions between these zones. The persistence of these clusters across independent LOYO runs points to regional biases rather than idiosyncratic county-level noise.

The strength of spatial clustering fluctuates within the leave-out year. The 2021 split shows the largest global clustering ($I=0.4908$), consistent with the visually exten-

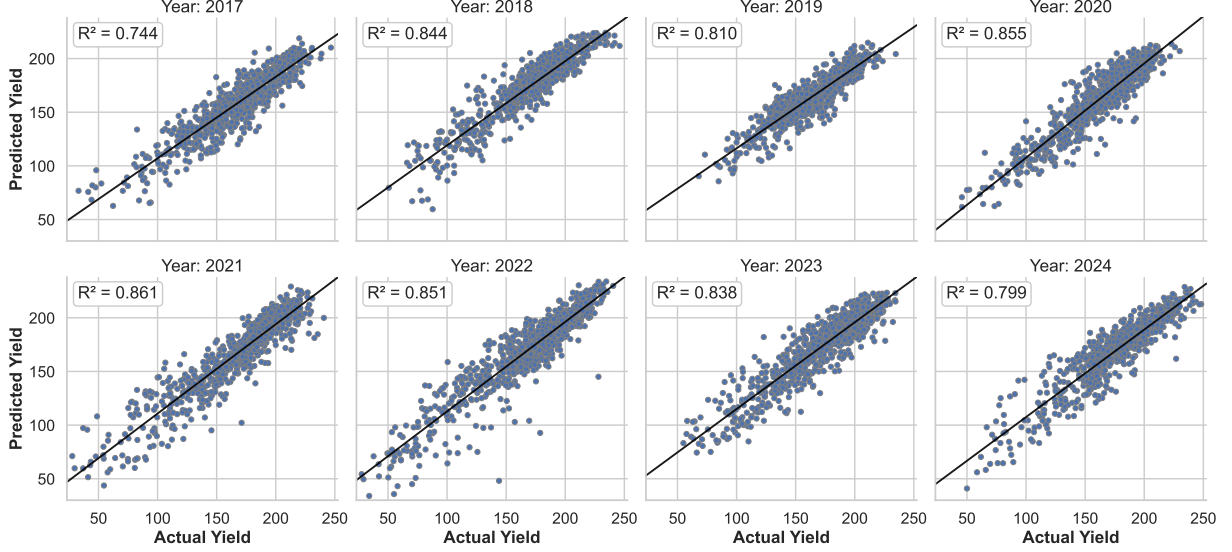


Fig. 2. Predicted vs. observed county-level corn yields under LOYO; points are counties from the held-out year; line shows $y=x$.

sive red HH area that year. In contrast, 2019 and 2024 exhibit the weakest (yet still strong) clustering ($I=0.3231$ and 0.3171), with more balanced mosaics of modest positive and negative errors. These differences likely reflect year-specific conditions interacting with regionally varying features that are only partially encoded by the static embeddings.

Collectively, the maps and statistics show that AEF embeddings alone capture a substantial share of spatial structure in yields, resulting smooth, geographically coherent residuals. The embeddings encode broad environmental and management regimes but may still miss some local or year-specific factors even though pretrained on large temporal dataset. Thus the bias has two components: (i) a global year effect shifting predictions slightly up or down, and (ii) regional over-/under-prediction that recurs across neighboring counties.

C. Dimensionality Analysis

The PCA decomposition reveals a moderately concentrated variance structure in the embedding space (Fig. 4a). The first principal component (PC) captures 22.1% of the total variance, with PC2 and PC3 contributing 18.0% and 14.5%, respectively. Cumulatively, the first three components explain 54.6% of the embedding variance, while the first 13 components reach 90%. Fig. 4a also demonstrates a gradual decline in variance explained. This long tail distribution suggests that while dominant patterns exist in the embeddings, agricultural information is distributed across multiple orthogonal dimensions rather than being confined to a low-dimensional manifold.

Despite capturing substantial embedding variance, the principal components showed limited ability to explain the variance in model prediction errors (Figure 4b). The largest univariate per-PC test R^2 values are small (on the

order of 2% - 4%), with no single PC accounting for a large share of performance variance. Most PCs have near-zero mean contribution. Cumulative multivariate test R^2 reached only 2.28% using the first 5 PCs and peaked at 2.63% with 21 PCs. This disconnect between variance concentration and error prediction indicates that variance-dominant directions do not correspond to error-relevant directions, a critical distinction often overlooked when interpreting PCA results.

D. Feature Importance and Explainable AI

Permutation importance analysis on held-out test years revealed a highly concentrated importance distribution among the embeddings. Fig. 5 a shows the Top-12 embedding dimensions by mean ΔR^2 across years (error bars indicate inter-year SD), where Fig. 5 b reports year-wise stability for the same set. The top five embeddings (A17, A38, A01, A28, and A49) collectively accounted for about 30% of the total importance share. A17 alone contributes 8.6%. Across all features and years, per-fold ΔR^2 values typically fall in the 0.03-0.10 range and peak at 0.137, indicating that *no* single dimension is solely responsible for generalization.

Importances are reasonably stable across years for a subset of features. A49 and A60 show low inter-year variability (coefficient of variation ~ 0.21), with values clustered around 0.05-0.06 and 0.03-0.05, respectively (Fig. 5 b). Others exhibit clear year effects: eg., A17 is consistently strong from 2018-2020, while A42 is weakest in 2017 but becomes informative thereafter. Directional associations suggest that higher values along A49, A34, A08, A42, and A60 tend to increase predicted yield, whereas increases along several other bands (A17, A24, A04, A28, A01 and A36) tend to decrease it; we report these signs

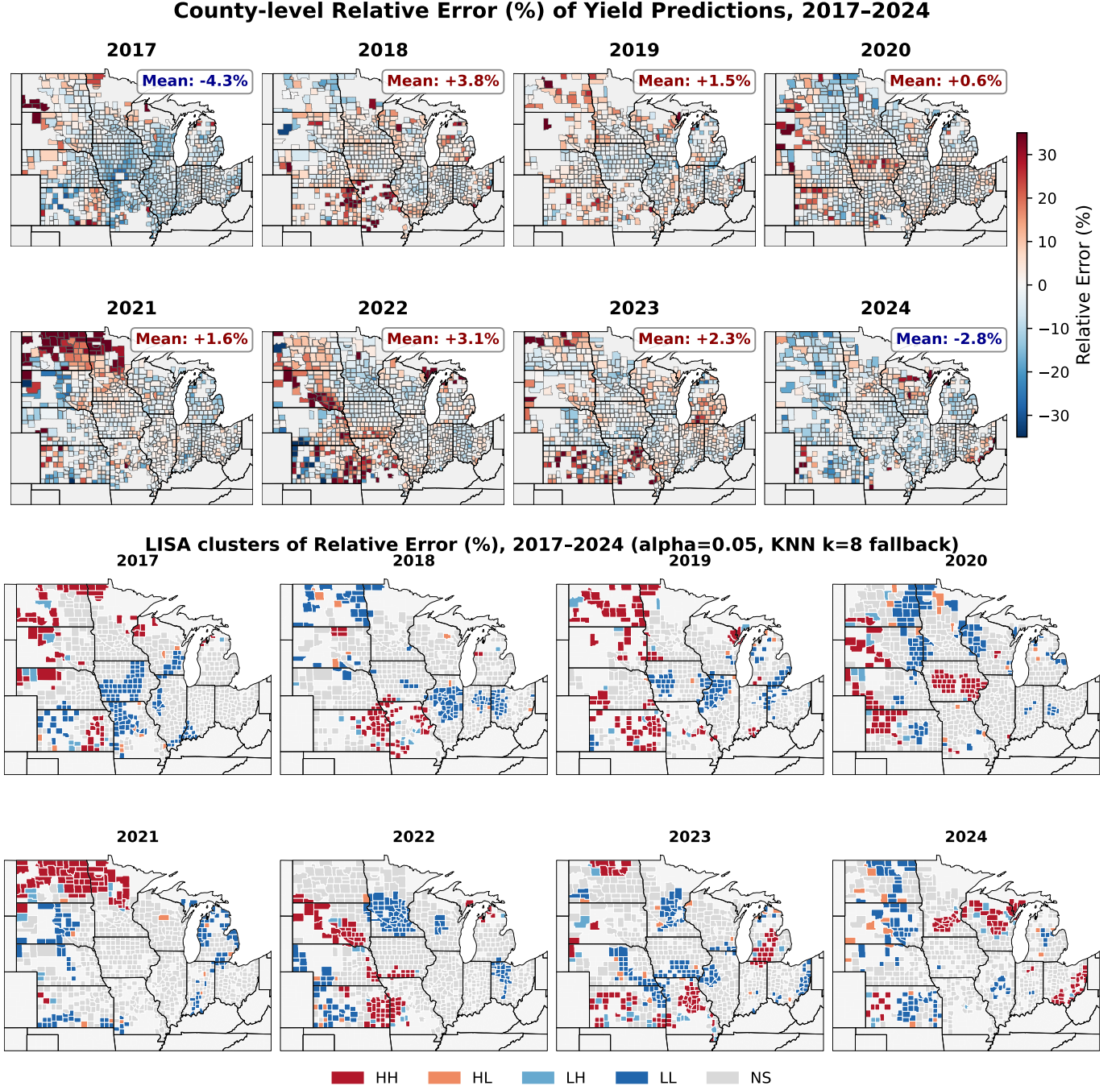


Fig. 3. (top) County-level relative error (%) under LOYO for corn. Blue = under-prediction; red = over-prediction. (bottom) LISA clusters (HH, HL, LH, LL) of relative error (%); $\alpha = 0.05$; $k = 8$ nearest neighbors.

as model-implied trends rather than causal effects (Figure III).

E. Label Efficiency

To quantify label efficiency, we held 2021 as the test year and randomly subsampled the training set to $\{1, 5, 10, 20, 30, 60, 90\}$ % of available counties (across the remaining years), repeating each fraction with 10 independent random draws. We report mean R^2 and its standard deviation over subsamples.

Performance rises sharply with a small number of labeled counties and then saturates (Fig. 6): at 5% of training data (~ 278 county-years) we already obtain $R^2 = 0.762 \pm 0.022$; by 20% ($\sim 1,111$) R^2 reaches 0.835 ± 0.010 . Using only 30% of the training labels achieves $R^2 = 0.854$. The dispersion across random subsamples shrinks with sample size (Std drops from 0.0272 at 1% to 0.0057 at 90%), indicating stable generalization once a modest label budget is available. These results imply that the annual AEF embedding already carries most of the

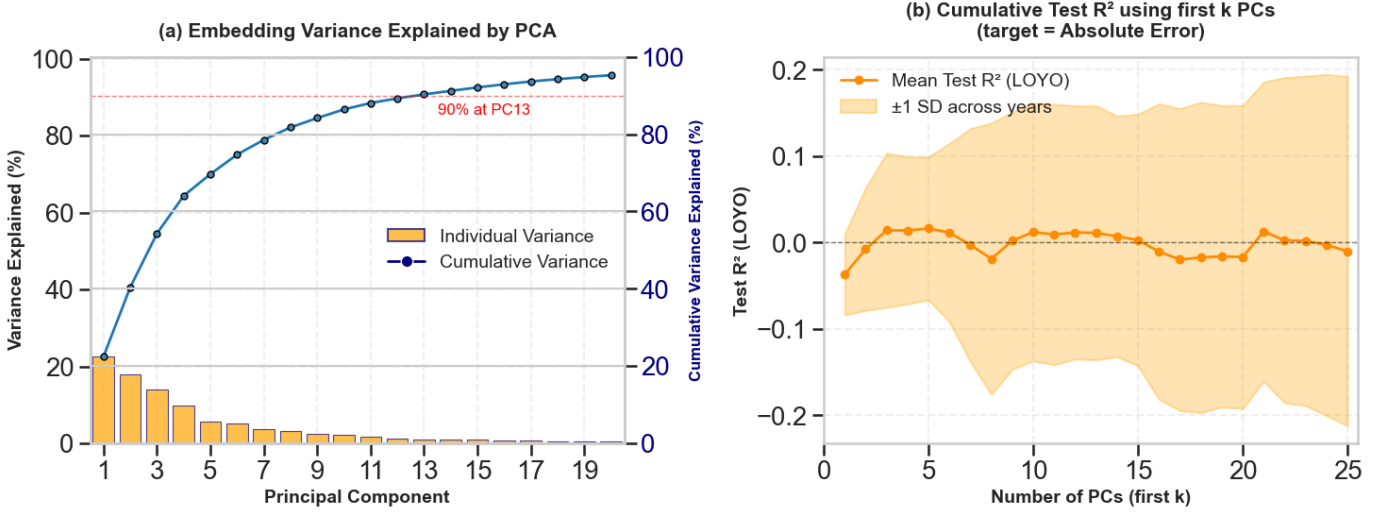


Fig. 4. (a) Scree: variance explained by PCs of AEF embeddings (train-fold PCA, test projections). (b) Test R^2 for predicting absolute error from first k PCs—low throughout.

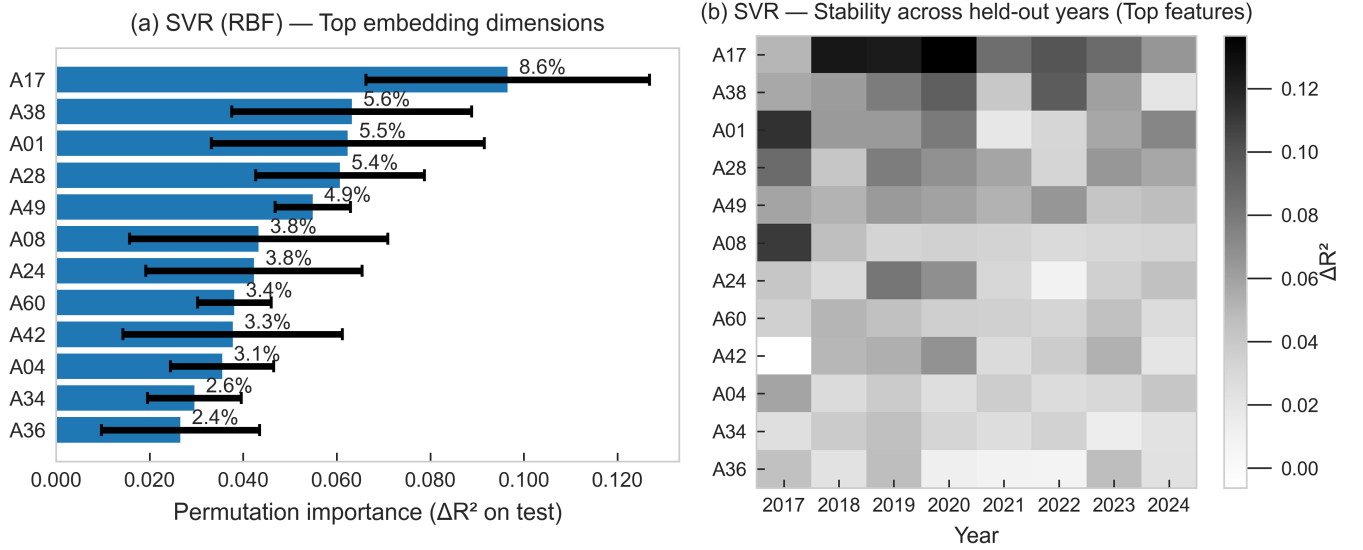


Fig. 5. Corn: (a) Top embedding dimensions by test-fold permutation importance (ΔR^2); error bars show inter-year SD. (b) Year-wise stability heatmap for the same features.

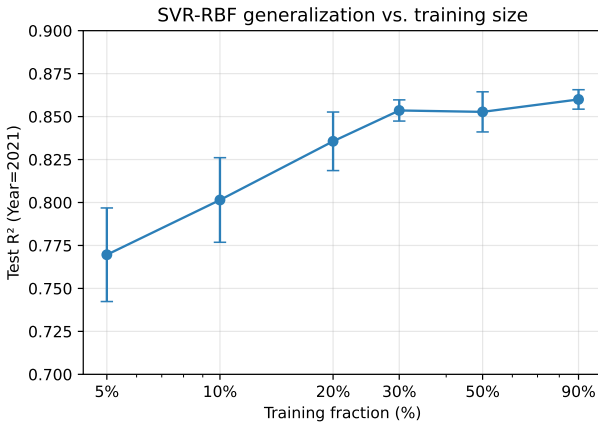


Fig. 6. Corn, test year 2021: test R^2 as a function of training fraction.

task-relevant variation, and the supervised layer mainly provides a light calibration.

F. Soybean: transferability and key differences

Using the identical pipeline and re-tuned RBF-SVR hyperparameters, soybean also attains strong out-of-year accuracy across 2017–2024 (mean $R^2=0.814$, RMSE 4.64 bu/ac; Table II). As with corn, performance peaks in 2021–2022 and dips in 2019, indicating similar sensitivity to inter-annual anomalies. AEF+SVR also outperforms both the county-mean and the NDVI+weather baselines in all eight folds, confirming that the embeddings carry crop-agnostic, inter-annual signal beyond fixed effects and seasonal summaries (Table II).

There are two major differences than prediction trends of corn. (i) Spatial analysis show *stronger* residual clus-

tering for soybean on average (higher global Moran’s I), implying regionally structured miss that is only partly encoded in the embeddings. (ii) Feature usage overlaps but is not identical: the most influential directions (e.g., A49, A17, A28) recur across crops, while a few axes (e.g., A24) flip sign (Table III), suggesting crop-specific semantics. See Appendix A (Figs. A–9) for residual maps, LISA, PCA error alignment, and permutation-importance details.

Finally, as in corn, principal directions of unsupervised variance explain little of the supervised error (soy PCs account for only a few percent of absolute-error variance), again indicating that predictive geometry lives in low-variance subspaces rather than dominant PCs (Appendix A, Fig. 7a).

TABLE III

OVERLAP OF THE TOP-10 EMBEDDING DIMENSIONS FOR CORN VS. SOYBEAN AND SIGN CONCORDANCE (DIRECTION OF EFFECT ON OUT-OF-SAMPLE PREDICTIONS).

Both Crops			Corn Only		Soybean Only	
Feature	Corn	Soy	Feature	Sign	Feature	Sign
A49	+	+	A38	–	A51	+
A17	–	–	A01	–	A03	+
A28	–	–	A08	+	A41	+
A24	–	+	A42	+	A05	–
			A34	+	A33	+
			A60	+	A62	–

IV. DISCUSSIONS

A. Geo-Foundation Embeddings as Analysis-Ready Features

A single, annual 64-dimensional embedding from a Geo-Foundation Model [13] functions as an analysis-ready feature set for county-scale yield estimation. Using only county-mean, crop-masked embeddings and a modest nonlinear regressor, we obtain strong, temporally out-of-sample skill under leave-one-year-out validation (2017–2024). For corn, the average test performance is $\bar{R}^2=0.825$ with RMSE 14.8 bu/ac ($\approx 8.8\%$ RMSE; Table II), and the best year reaches $R^2=0.861$. Model sweep confirms that modest nonlinearity matters: RBF-SVR consistently outperforms linear and tree baselines (Table I), indicating that yield responds to a smooth, low-dimensional manifold in the 64-D embedding space.

Operationally, representation learning has absorbed much of the engineering traditionally required for index design, time-series summarization, and weather feature construction. Because AEF embeddings compress multi-sensor observations as well as other spatio-temporal context into a compact vector, the downstream pipeline reduces to (i) crop masking, (ii) county averaging in embedding space, and (iii) a small supervised model. Although training-year averaged baseline is already very strong (mean $R^2\approx 0.69$), AEF reduces RMSE by $\sim 23\%$ (corn) and $\sim 22\%$ (soybean) on average. The NDVI+weather stack lags AEF particularly for corn, consistent with NDVI saturation [38], [39] and the limits of monthly precipitation summaries to capture event-scale stress. The fact that

AEF wins by large margins supports the view that the representation already encodes much of the subtle phenology and moisture/structure cues than many hand-crafted indices.

Learning-curve experiment with 2021 as the held-out year shows that the AEF representation is highly label-efficient, with 30% of labels the model attains $R^2=0.854$, within 0.009 absolute of the 90% setting (0.861; Fig. 6). The same workflow transfers to soybean without architectural changes, achieving $\bar{R}^2\approx 0.81$ across the same years (Table II), which supports the view that the embedding captures crop-agnostic environmental and phenological structure.

A forecasting-style roll-forward test yields essentially the same accuracy as LOYO with negligible bias difference. This parity suggests that AEF embeddings support stable mappings that do not rely on information from future seasons, strengthening the case for operational deployment.

B. Spatial Intelligence in Embedding Space

County-level errors are geographically organized. Global Moran’s I on relative errors is positive and highly significant in every held-out year for both crops, with corn ranging $I=0.317$ – 0.491 (mean $\bar{I}\approx 0.387$; Table IV) and soybean $I=0.264$ – 0.581 (mean $\bar{I}\approx 0.484$; Table IV). LISA maps (Fig. 3b) highlight recurrent HH/LL clusters that persist across years and align with broad agro-ecological gradients. This coherence arises even though the model does not ingest coordinates or explicit spatial covariates, implying that the embeddings themselves carry latent geographic context (e.g., soils, moisture regimes, management intensity) that varies smoothly across space.

In future practice, the small year-level bias combined with spatially structured residuals suggests a simple, interpretable corrective: a year intercept plus a low-complexity spatial adjustment (e.g., county-level bias mapping or geo-statistical smoothing) can likely eliminate most remaining error without complicating the upstream feature pipeline [40]. In addition, the spatial diagnostics are actionable: persistent HH or LL clusters flag regions where unmodeled drivers (e.g., cultivar groups, irrigation prevalence, late-season stress) systematically tilt predictions. These regions are prime candidates for targeted ground-truthing or for introducing a minimal set of auxiliary variables if operational constraints allow.

C. Dimension Parsimony and Latent Semantics

Explainability analyses indicate that predictive signal is concentrated but not single-feature dominated. Test-fold permutation importance shows that a small subset of AEF dimensions repeatedly carries out-of-sample leverage (Fig. 5a–b); several dimensions (e.g., A49, A60) are stable across years, while others (e.g., A17, A01, A24) exhibit clear year effects. Cross-crop comparisons reveal a shared core—A49, A17, and A28 rank highly for both corn and soybean—while A24 flips sign between crops (Table III),

pointing to crop-specific semantics along some embedding axes (e.g., differences in canopy structure, residue, or background soil exposure). These signs reflect model-implied associations on held-out data, not causality, but they provide a compact qualitative summary of how the regressor uses the representation.

Principal directions of unsupervised variance in the embeddings do not align with supervised error structure. Predicting absolute error from the first k PCs explains only a few percent of variance even at moderate k (corn $< 3\%$, soybean $< 4\%$; Fig. 4), despite PCs capturing most embedding variance. This disconnect cautions against equating “variance explained” with “task relevance”: the predictive geometry appears to live in low-variance, task-discriminative subspaces rather than along dominant unsupervised axes. This explains strong LOYO stability: the model captures broad, regular patterns rather than memorizing year-specific idiosyncrasies. In practice, this supports targeted dimensionality strategies guided by supervised importance (e.g., selecting the top k features or learning a small supervised rotation) over unsupervised truncation by PCA.

D. Limitations and Future Work

Our study is intentionally minimalistic: we only use annual AEF embeddings and a single supervised learner—so several limitations remain. First, the county-mean aggregation discards within-county heterogeneity (e.g., topography, soil properties, and management practices), which likely contributes to the spatially coherent residuals we observe. Second, AEF embeddings summarize a calendar year, limiting the temporal granularity we could control. Third, we do not include explicit exogenous drivers (e.g., seasonal precipitation anomalies), which could remove the small year-level intercept shifts we observe. Last but not least, even with ablation and permutation study, the physical semantics of individual embedding dimensions remain unknown. Linking directions (e.g., A49/A17/A28) to phenology, canopy water status, or soil context will require targeted experiments and ancillary data.

V. CONCLUSION

We evaluated whether annual 64-D embeddings from AlphaEarth Foundations can function as analysis-ready features for county-scale yield estimation. Using only crop-masked county means and a modest support vector regression, the model achieved strong out-of-year generalization under leave-one-year-out across 2017–2024 for both corn and soybean, demonstrating that representation learning can subsume much of the traditional index design and feature engineering. Residuals are modest in magnitude yet spatially coherent, implying that the embeddings already encode broad agro-ecological structure while leaving small, correctable components (e.g., a year intercept plus a low-complexity spatial adjustment). Explainability analyses show that predictive leverage concentrates in a

relatively small subset of embedding axes, whereas principal directions of unsupervised variance are poor proxies for task relevance. The pipeline is therefore lightweight (*mask-mean-regress*), reproducible, and competitive with more complex alternatives, while transferring across crops without architectural changes. Future work will quantify simple residual corrections, explore pixel-level or bag-level formulations to recover within-county heterogeneity, and link salient embedding directions to physical factors (phenology, canopy water status, soils), enabling interpretable, geographically scalable agricultural monitoring.

APPENDIX A

SOYBEAN SUPPORTING RESULTS

This appendix provides diagnostics for soybean that parallel the corn analysis while avoiding repetition in the main text. We report spatial error structure, local clustering, unsupervised variance geometry, and supervised feature usage, complementing the main soybean metrics in Table II.

Spatial Pattern of Errors

Figure 9a maps county-level relative errors (%) for each LOYO split. As in corn, errors are geographically coherent rather than speckled, with contiguous bands of over- and under-prediction that track agro-ecological gradients. Year-level mean bias is small in most years; 2023 shows a larger positive uptick (+6.6%). Across 2017–2024 the average percent RMSE is 9.34% (Table II), indicating modest, agronomically reasonable uncertainty at county scale.

Global and Local Spatial Autocorrelation

Soybean residuals are more clustered than corn on average. Using the same $k=8$ nearest-neighbor weights, global Moran’s I ranges from 0.2639 (2020) to 0.5807 (2017), with mean $\bar{I} \approx 0.464$ and $p_{\text{perm}}=0.001$ in all years (Table IV). Figure 9b shows LISA clusters (HH/LL/HL/LH) at $\alpha=0.05$: repeated HH zones appear along parts of the northern tier in several years, while LL patches recur in portions of the central Corn Belt, with HL/LH edges at their boundaries. The consistency of these patterns across independent LOYO folds suggests missing slow-varying drivers (e.g., rotation/management regimes or soils) rather than idiosyncratic noise.

Unsupervised Embedding Geometry vs. Error

The soybean embeddings exhibit a moderately concentrated variance structure: PC1 explains $\sim 23.3\%$ of variance; PCs 1–3 explain $\sim 54.6\%$; and 90% is reached by PC 12 (Fig. 7a, left). However, these dominant directions explain little of the supervised miss. Predicting absolute error from the first k PCs yields low test R^2 throughout (maximum $\sim 4.05\%$ at $k=24$; Fig. 7a, right). Thus, the geometry relevant for generalization lives in lower-variance subspaces rather than along variance-dominant axes, mirroring the corn finding.

TABLE IV

GLOBAL MORAN'S I OF COUNTY-LEVEL RELATIVE ERROR UNDER LOYO (COUNTY $k=8$ NEAREST NEIGHBORS; 999 PERMUTATIONS).

Year	N	I	$E[I]$	z	p_{perm}
2017	836	0.4361	-0.0012	26.26	0.001
2018	737	0.4585	-0.0014	26.02	0.001
2019	681	0.3231	-0.0015	17.66	0.001
2020	932	0.3300	-0.0011	16.20	0.001
2021	794	0.4908	-0.0013	28.77	0.001
2022	859	0.3813	-0.0012	23.25	0.001
2023	811	0.3560	-0.0012	21.12	0.001
2024	700	0.3171	-0.0014	17.60	0.001

(a) Corn

Year	N	I	$E[I]$	z	p_{perm}
2017	803	0.5807	-0.0012	25.56	0.001
2018	718	0.4852	-0.0014	27.12	0.001
2019	640	0.4232	-0.0016	22.44	0.001
2020	882	0.2639	-0.0011	16.32	0.001
2021	754	0.5755	-0.0013	32.96	0.001
2022	829	0.3994	-0.0012	23.93	0.001
2023	758	0.5753	-0.0013	33.04	0.001
2024	658	0.4053	-0.0015	21.90	0.001

(b) Soybean

Test-Fold Feature Importance and Stability

Permutation importance computed *only* on held-out data (20 shuffles per feature per fold) reveals a concentrated but not single-feature-dominated profile (Fig. A). Three of the most influential directions (A49, A17, A28) overlap with corn, with A49 ranking first ($\Delta R^2 \approx 7.2\%$) and A17 second ($\sim 5.6\%$). Soybean elevates additional axes (A51, A03, A41, A05) that were not top-ranked for corn. As a set, the top-12 features account for $\sim 45\%$ of the summed test-fold importance, indicating that a relatively small subset of the 64 channels carries most of the out-of-sample leverage. Year-wise stability heatmaps (Fig. A) show A49 informative in most splits with low dispersion; A17 is especially strong from 2018–2020; and A42 tends to be weak early and more informative later.

Figure 8 reports the Pearson correlation between each embedding dimension and the *out-of-sample* SVR predictions pooled across folds (sign only; not causal). Signs largely agree with corn for shared leaders (A49 positive; A17 and A28 negative), while A24 flips sign (negative in corn, positive in soybean), suggesting crop-specific semantics along that axis.

Summary. (i) Soybean achieves strong accuracy with the same embedding-only pipeline; (ii) residuals are geographically organized and *more* clustered than corn, pointing to slow-varying regional factors; (iii) dominant PCA directions do not explain error; and (iv) a compact subset of AEF channels repeatedly drives generalization, with a shared core across crops and a few crop-specific axes.

ACKNOWLEDGMENT

The authors acknowledge the USDA NASS Quick Stats database and Google Earth Engine for data access, and thank the reviewers for their constructive feedback.

CODE AND DATA AVAILABILITY

GEE codes to reproduce county-level embedding data will be released. Jupyter notebooks to reproduce all table and figures, including modeling process, will be released upon acceptance.

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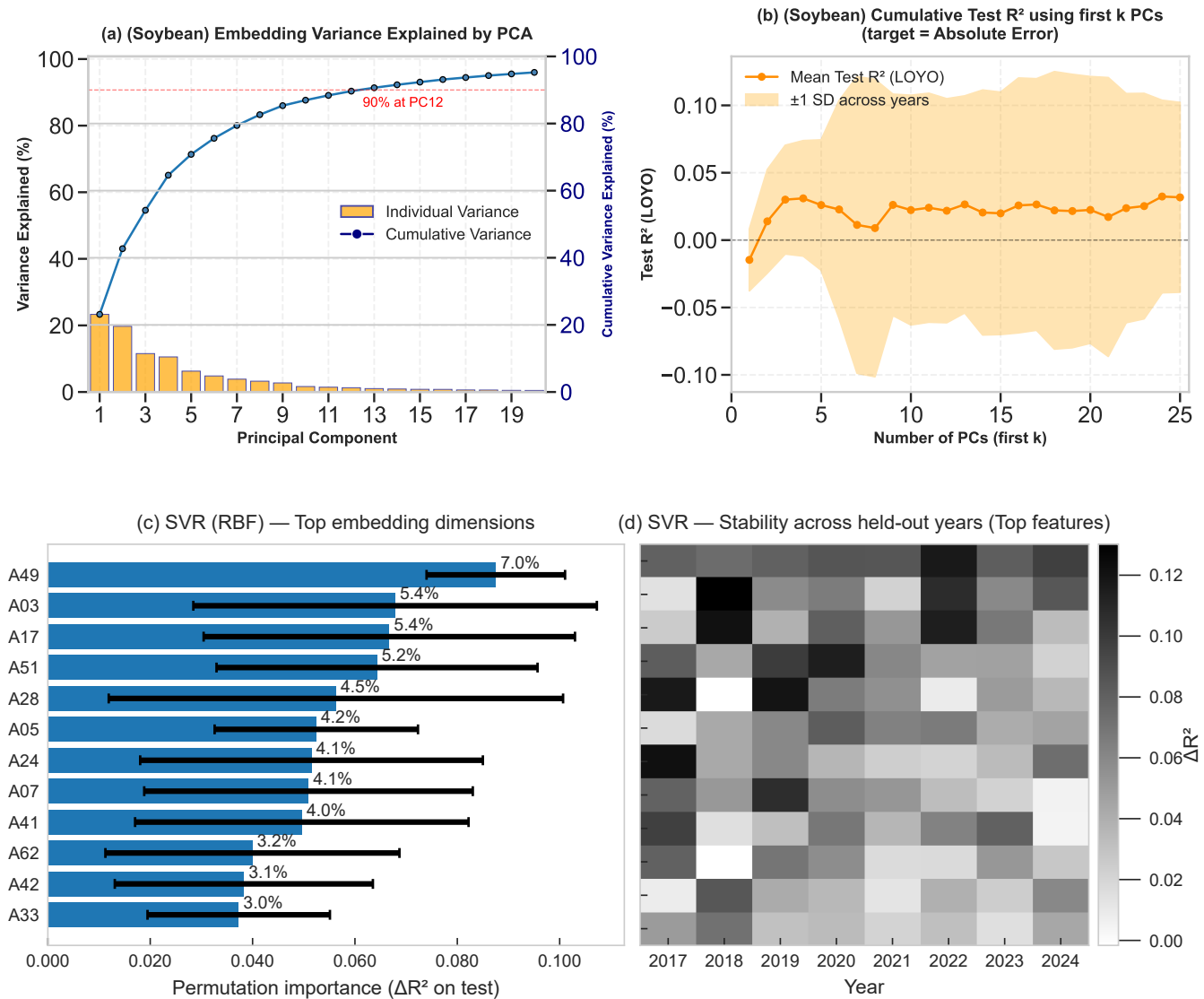


Fig. 7. Soy: (a) Scree: variance explained by PCs of AEF embeddings (train-fold PCA, test projections). (b) Test R^2 for predicting absolute error from first k PCs—low throughput. (c) Top embedding dimensions by test-fold permutation importance (ΔR^2); error bars show inter-year SD. (d) Year-wise stability heatmap for the same features

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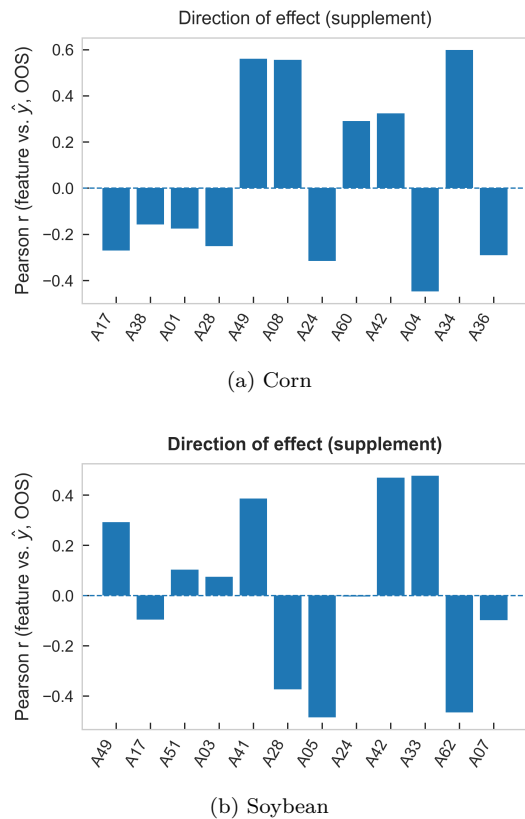


Fig. 8. Direction of effect: Pearson correlation (r) between each AEF feature and out-of-sample SVR predictions, pooled across folds.

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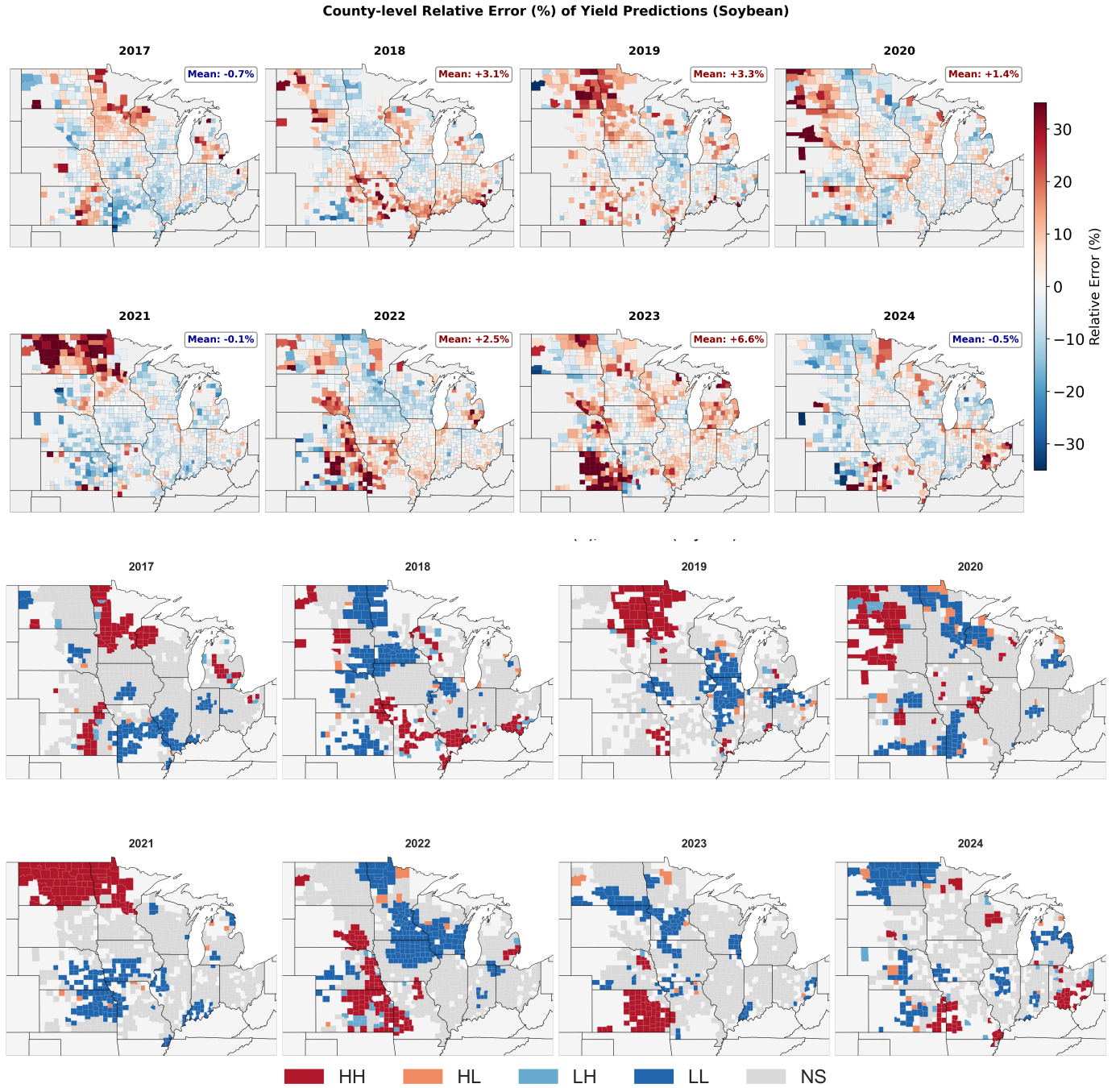


Fig. 9. Soybean: (top) County-level relative error (%) under LOYO for both crops. Blue = under-prediction; red = over-prediction. (bottom) LISA clusters (HH, HL, LH, LL) of relative error (%); $\alpha = 0.05$; $k = 8$ nearest neighbors.